

Application of blockchain for secure data transmission in distributed state estimation

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Abstract—The application of renewable energy sources in the power grid increases the necessity of tracking the system's state, especially in smart grids, where there is a bidirectional transfer of data and power. The complexity of coupling between communication and the electrical infrastructure in a smart grid will create a higher chance for security breach. Increasing the state estimation accuracy will help the smart grid operator efficiently manage the system. The paper proposes an integration of distributed state estimation with a blockchain designed communication platform. Additionally, the asynchronous manner for data transmission, which is more likely to happen in the real world, has been considered as the second task of this research. Finally, a detailed analysis of the blockchain-based application in distributed state estimation is provided. The numerical analysis shows that the proposed method meets real-world performance requirements and brings high security and reliability to the distributed state estimation process.

Index Terms—Blockchain, Distributed algorithms, Optimization, Power system control, Smart contract, State estimation

I. INTRODUCTION

POWER grid has faced different challenges due to the increasing utilization of the distributed energy sources, and the non-stop escalating level of energy demand [1]. Hence, state estimation (SE) plays an essential role in justifying and regulating system operator decisions like economic dispatch, load frequency control, electricity markets, and load forecasting. It is to be noted that advanced SE can improve monitoring and controlling the power grid in case of a contingency. Especially for the smart grids in which bidirectional transfer of electrical energy and system/consumer data increases the complexity [2], [3]. Such a system can be divided into two integrated parts, i.e., physical equipment of a traditional power system (Physical part), and telecommunication equipment (Cyber part). Combining these two parts will lead to a cyber-physical power system (CPPS) [4].

A. Power system state estimation

Although SE is highly comparable to the conventional load flow, it considers the unpredictable errors that might originate

due to unexpected system changes, meters or communication system errors, inaccuracy in equipment calibration, planned manipulation from a malicious attacker, etc. [3], [5]. Additionally, conventional load flow analysis does not consider redundancy and imprecision of the system's measurement data, whereas SE considers the mentioned features [5].

Considering a brief background of the evolution of the SE method and its application in power system, it is worth noting that as soon as Schweppe pointed out the application of SE in power system, it attracted industrial communities' attention [6]. However, they did not apply the basic weighted least squares (WLS) method proposed by him at the beginning but utilized two different methods, developed by Dopazo [7], and Larson [8]. However, after a while revised version of Schweppe's method was admitted, and those two methods were excluded from industrial applications [9]. Taking into account the sparsity of the gain matrix, different researchers made several attempts to slightly improve the WLS estimator [10]. Also, SE related issues like ill-conditioning appeared afterwards, and several methods such as the inclusion of zero injection power within constraints of SE were proposed to overcome these issues [2], [11], [12]. Schweppe et al. proposed applying non-quadratic estimators for bad data detection within the early years. Later on, there were plenty of studies on methods for identifying bad data, such as application of least absolute value [13]. It is to be noted that from the early stages of utilizing SE, the mentioned areas and problems have been an active research area for the power systems, operations and control research community.

Growth of the power system due to increase in the level of needed electricity consumption and propagation of the communication technology, bring in several problems assigned with power system operation, especially centralized SE (CSE) such as *Expansion of power system continent-wise, Policy and privacy, Dimension of the grid, Communication bottleneck, Data size and Security/Reliability*, to name a few.

Expansion of the power grid over continents makes an interconnected system that these continents can be affected by contingencies in other ones [14]. Although, in some regional expansion cases, e.g. the case with regional transmission organizations (RTOs) in Europe, operators are using HVDC technology for power transfer which is also another research area for considering hybrid HVDC/AC SE so that they can meet the characteristics of the new network regarding Supervisory Control and Data Acquisition (SCADA) system [15]. Vulnerability and inflexibility of CSE makes it unsuitable for a multi-area (or multi RTO) estimator from policy and

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